Kaiwen Mo

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EDUCATION

University of California, Los Angeles - *Master of Science in Computer Science* *Expected May 2028*

University of Michigan – Ann Arbor **GPA: 3.80 / 4.00**
Bachelors of Science in Computer Science and Economics Honors *May 2026*

SKILLS

Programming & Web: *C++, Python, JavaScript, SQL, React, Flask, HTML/CSS, Git*
Systems & Backend: *Distributed Systems, MapReduce, REST APIs, AWS EC2*
Applied ML & Security: *OpenCV, YOLOv8, MediaPipe; Wireshark, Disk Forensics (Autopsy), Web (SQLi, XSS, CSRF)*

WORK EXPERIENCE

Wade Trim (Multidisciplinary Design Program - University of Michigan) **Ann Arbor, MI**
Software Engineer Intern (Industry-Sponsored) *Jan. 2026 – Present*

- Architecting a full-stack Digital Twin platform using React and Leaflet to synchronize GIS-integrated simulation data (PCSWMM) with real-time interactive infrastructure modeling.
- Collaborating with Wade Trim engineers to build scenario-based simulation workflows and optimized data pipelines, reducing simulation-to-visual latency for complex urban datasets.

Next Play Games Inc. **Remote**
Software Engineer Intern | AI Team Lead *May 2025 – Aug. 2025*

- Designed and deployed a real-time video analytics pipeline integrating computer vision inference and rule-based event logic for live football broadcasts.
- Architected modular backend services to project player/ball positions into a unified field coordinate system, algorithmically inferring plays such as passes and touchdowns.
- Implemented a second-stage event detection service to classify advanced game events (fumbles, flags), significantly improving system robustness beyond raw model predictions.
- Developed SportLingo, a full-stack training application using MediaPipe pose tracking to provide real-time feedback for gamified youth sports movements.

PROJECT EXPERIENCE

MapReduce Framework - Distributed Systems *Feb. 2025 – Apr. 2025*

- Designed and implemented a distributed MapReduce execution framework with a centralized Manager and multiple Worker processes, enabling parallel execution of map and reduce tasks across large datasets.
- Built socket-based inter-process communication and multi-stage data pipelines (map → group → reduce) for worker registration, task scheduling, and result aggregation, ensuring correctness and coordination under concurrent workloads.

Scalable Search Engine - Information Retrieval *Apr. 2025 – Apr. 2025*

- Built a scalable search engine by constructing a segmented inverted index over large document collections.
- Implemented relevance ranking and query serving by integrating tf-idf and PageRank scoring with RESTful backend APIs that return ranked JSON and power a frontend search interface.

Stock Market Simulation (C++, Deployed Web Interface) *Oct. 2024 – Nov. 2024*

- Designed a C++ simulator for an electronic stock exchange capable of processing 100,000+ trades across 20 traders and 25 stocks within seconds, incorporating a 'time-traveler' feature to explore optimal trading strategies.

Real-Time Social Media Platform *Jan. 2025 – Mar. 2025*

- Designed and built a full-stack web app using React, Flask, and AWS EC2, supporting features like infinite scrolling, real-time updates, and double-click to like; optimized for 100+ concurrent users.
- Implemented a SQL database and REST API for user data management, including robust user authentication and session handling with Flask cookies, employing skills in cloud deployment and full-stack integration.